

SPRING BOOT

CHAPTER – II

@SpringBootApplication:

- @SpringBootApplication is the main annotation used to start a Spring Boot application.
- It is placed on the main class of the program.
- @SpringBootApplication is a combination of @Configuration, @EnableAutoConfiguration, and @ComponentScan used to start and configure a Spring Boot application.

@Configuration:

- Marks the class as a source of bean definitions. It tells Spring that this class contains configuration for the application.

@EnableAutoConfiguration:

- Spring Boot automatically configures the application based on dependencies in pom.xml.

@ComponentScan:

- Spring automatically scans packages for components like:
 - @Component
 - @Service
 - @Repository
 - @Controller
 - @RestController

Annotation	Purpose
@Configuration	Defines beans
@EnableAutoConfiguration	Automatic configuration
@ComponentScan	Scans components

Introduction to Spring Framework:

<code>ApplicationContext</code>	Spring container
<code>getBean()</code>	Get object from container
<code>@Component</code>	Tell Spring to create object
<code>Singleton</code>	One object only
<code>Prototype</code>	New object every time

ApplicationContext (Spring):

- ApplicationContext is the main container of the Spring Framework.
- It is responsible for:
 - Creating objects (called Beans)
 - Storing those objects
 - Managing them
 - Providing them when the program needs them

@Component:

- @Component is a Spring annotation used to mark a class as a “Spring-managed bean,” meaning Spring will automatically create, manage, and inject its object wherever needed.

@Scope:

- @Scope is used to define how many objects (instances) of a bean Spring should create and how long they live inside the Spring container.
- Spring mainly uses two scopes:
 - Singleton
 - Prototype

SINGLETON SCOPE:

- Singleton means only one object is created.
- Even if we call `getBean()` many times, the same object is returned.
- It is a Default Scope

PROTOTYPE SCOPE:

- Prototype means Spring creates a new object every time.
- A new instance of the bean is created every time `getBean()` is called.

Visual Representation

